

Relevant Reading

- [Flight Vehicle Design Book](#)
 - Chapter 2: Fundamentals (13 pages)
 - Chapter 6.3: Propellers (4 pages)

Problem 1 Vocabulary

Explain the following terms; making sure to use sufficient detail, including any math or helpful figures. In some cases, these terms are simple one sentence definitions, in others, you should include several paragraphs to explain them fully.

Airfoil Geometry Terms

- Chord
- Leading Edge
- Trailing Edge

Rotor Geometry

- Hub Radius
- Tip Radius
- Twist
- Pitch
- Precone
- Sweep

Non-dimensional Numbers

- Reynolds Number
- Mach Number
- Advance ratio
- Tip Speed Ratio
- Solidity

Forces and Moments

- Lift
- Drag
- Pitching Moment
- Lift and Drag Polars
 - Angle of Attack
 - Zero Lift angle of attack
 - Lift Curve Slope
 - Stall

Problem 2 Exploration

Complete the following exploration.

2.a Prerequisites

- i. [Install CCBBlade.jl](#)
- ii. Complete the [Quick Start Tutorial](#)

2.b Load Distributions and Rotor Efficiency

Blade Element Momentum Theory (BEMT) allows for the analysis of propellers, wind turbines, and helicopter blades. Choose one that interests you and implement it in CCBBlade.jl.

- i. Plot the section loads (normal and tangential loads) vs radius. Analyze and explain the shape of the load distribution. Why does the loading increase? Why are there decreases?
- ii. Compare the thrust, torque, and root bending moment vs varying rotor tip radius.
- iii. Compare the thrust, torque, and root bending moment vs varying the rotor section chord length.